PRIYANSHU MISHRA

Software Engineer | Full-Stack | BSc Computer Science, UNSW Sydney
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PROFESSIONAL SUMMARY

Computer Science graduate from UNSW with Distinction and experience building full-stack, backend and data-intensive systems. At ArmsOA, developed a real-time network resilience platform using Python/FastAPI, React, TypeScript and Docker, delivering production APIs, operator-facing features, simulation logic and observability backed by 500+ regression tests. Previously built end-to-end research software for encrypted network traffic analysis, spanning data pipelines, browser integration and machine learning. Strong across backend engineering, full-stack development, API design, testing and reliable software delivery.

EDUCATION

Bachelor of Science (Computer Science) · UNSW Sydney · Distinction WAM (78) *Sep 2023 – Aug 2026*
Coursework: Software Engineering, Data Structures and Algorithms, Object-Oriented Programming, Machine Learning

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, C++17, C, SQL, Java
- **Backend:** FastAPI, REST, OpenAPI 3.0, JSON Schema, WebSockets, Redis, Docker
- **Frontend:** React, TypeScript, Next
- **Testing and delivery:** automated regression testing, GitHub Actions CI, Git
- **AI and ML:** LLM agents (Claude Opus/Sonnet), PyTorch, scikit-learn, Pandas, NumPy, OpenCV
- **Systems and networks:** POSIX processes and signals, multithreading, UDP/TCP sockets, Scapy, tshark, AddressSanitizer, web and binary exploitation

EXPERIENCE

Software Engineer · ArmsOA, Canberra *Jun 2026 – Sep 2026*

- Built backend services and the operator interface for a real-time network resilience simulator (Python/FastAPI, React, TypeScript, Docker), modelling nodes, multi-link edges and prioritised bearers across 4G/5G, microwave, SATCOM and UHF/VHF under failure and environmental stress.
- Wrote the contract layer (OpenAPI spec, versioned JSON schemas, AI interface contract) that frontend, backend and AI work were built against, and labelled mock endpoints as mock so unfinished capability never looked real.
- Integrated Claude Opus/Sonnet agents into the product with live-state context, persistent sessions and an operator approval step before any recovery action runs.
- Implemented deterministic multi-run recovery simulation with per-candidate confidence scoring, kept physical-link failures auditable and separate from restored state, and added JSON/GeoJSON exports. Turned client specifications into production APIs backed by 500+ backend regression tests.

AI Researcher · UNSW School of Electrical Engineering, Sydney *Jun 2025 – Feb 2026*

- Built a Firefox WebExtension that intercepts live DoH requests, a Python/Scapy pcapng sniffer and a Selenium harness that correlates encrypted traffic with browser sessions.
- Built the ingestion and feature pipeline and an ML classifier separating DoH/DoT/DoQ from plaintext fallback, reaching 92% accuracy on real captures.
- Owned the work end to end, from scoping to evaluation, and presented findings to engineers and non-technical infrastructure stakeholders.

Resident Advisor · Shalom College, UNSW, Sydney *Feb 2026 – Present*

- First point of contact for 130 residents: on-call duty, welfare checks and confidential incident escalation.

PROJECTS

Midpoint Fuzzer | C++17, POSIX, AddressSanitizer

- Built a multithreaded C++17 black-box fuzzer with a fork/exec harness, poll-based timeouts, signal-level crash classification and SHA-256 crash deduplication, driven by three format-aware mutators (recursive JSON AST, CSV and byte-level). Children run under setrlimit memory caps with ASan/UBSan support.

Quant AI Live Intelligence | Python, FastAPI, TypeScript, Redis, Docker

- Turns a company name into a forecast from price history, news and Reddit via documented APIs. Handles outages and rate limits explicitly (clear 429s, no fabricated data), with Redis caching and CI that runs the full suite without live API keys.

Tributary | Java, REST APIs, Event Driven Architecture

- Built a distributed messaging platform with message streaming, partitioning and consumer groups, designed API-first with modular OOP, unit tests and documented interfaces.

ACHIEVEMENTS

- UNSW Taste of Research Scholarship (competitive grant) to lead original research in AI agents for network observability
- Top 1% nationally, STEM and English Olympiads; State Topper in Science (IAIS, UNSW); High Distinction, Australian Chemistry Quiz (RACI)